

Data Acquisition, Cleaning, and Merging Pipeline

This report documents the workflow and operations implemented in `download_data.py`, `clean_data.py`, and `src/merge_datasets.py` to fetch, extract, clean, and merge datasets for the melanoma immunotherapy response predictor.

Codebase Architecture & Helper Modules

To maintain a clean and modular architecture, helper functions are organised under the `src/utils/` package directory:

- `src/utils/io.py`: Handles low-level file I/O operations including:
 - File downloading with streaming (`download_file`).
 - Gzip extraction and decompression (`extract_tar_gz`, `download_and_decompress_gzip`).
 - Conditional download checks (`download_if_missing`).
- `src/utils/preprocessing.py`: Coordinates data cleaning, normalisation, and sample alignment, including:
 - TCGA sample ID standardization (`standardise_sample_id`).
 - Survival status mapping (`parse_survival_status`).
 - Clinical metadata filtering (`clean_clinical_df`).
 - RNA-seq sample/gene filtering (`clean_rnaseq_df`).
 - Sample alignment between clinical and expression matrices (`align_expression_and_clinical`).
- `src/utils/cbioportal_api.py`: Wraps the cBioPortal REST API v2 client and provides functions to programmatically query study metadata, molecular profiles, sample lists, and download raw datasets (`download_raw_tcga_skcm`).

1. Data Acquisition (`scripts/download_data.py`)

The data acquisition script standardizes the raw input files by downloading official, curated datasets from the **cBioPortal DataHub**. It processes four cohorts:

- Liu 2019** (`mel_iatlas_liu_2019`)
- Hugo 2016** (`mel_iatlas_hugo_ucla_2016`)
- Riaz 2017** (`mel_iatlas_riaz_nivolumab_2017`)
- TCGA-SKCM** (`skcm_tcga_pan_can_atlas_2018`)

Workflow Details

- Directory Initialisation**: Creates the necessary target directories under `data/raw/{study_name}`.
- Targeted Downloads**: Fetches tarballs (`.tar.gz`) directly from `https://datahub.assets.cbioportal.org`.
- Redownload Prevention**: Checks for the existence of `data_clinical_patient.txt` in the destination directory to skip files that have already been retrieved.
- Extraction & Clean-up**: Extracts the archive contents to a temporary folder, moves the files into the final destination folder, deletes any empty parent directories, and deletes the temporary `.tar.gz` archive to save disk space.

2. Data Cleaning (`scripts/clean_data.py`)

The data cleaning pipeline transforms raw inputs into normalized expression matrices and clinical metadata dataframes suitable for modeling.

Core Parsers and Helpers

- `parse_cbioportal_expression(expr_file_path)`: Consolidates common cBioPortal expression parsing steps:
 - Reads the expression file (TPM or RSEM values).
 - Excludes missing Hugo gene symbols.
 - Averages duplicate symbols.
 - Drops the administrative `Entrez_Gene_Id` column.
- `parse_maf_mutations(raw_dir, sample_ids)`: Parses mutation data from the MAF format file (`data_mutations.txt`):
 - Filters for non-silent (non-synonymous) variants in `BRAF`, `NRAS`, and `NF1` genes.
 - Generates binary indicator variables (`mut_BRAF`, `mut_NRAS`, `mut_NF1` set to `1` if mutated, `0` otherwise) to join with clinical metadata.

Cohort-Specific Transformations

A. Liu 2019 (`clean_liu_2019`)

- Merges patient and sample clinical sheets.
- Maps response (`RESPONSE`) to binary target variable `response` (`CR/PR` → `1`, `PD` → `0`) and standardises clinical metadata (`patient_id`, `os_months`, `os_status`, `age`, `gender`).
- Log2-transforms TPM expression matrix ($\log_2(\text{TPM} + 1)$) and aligns samples.

B. Hugo 2016 (`clean_hugo_2016`)

- Merges sample and patient clinical sheets.
- Maps response outcomes to binary target `response`.
- Maps sex/gender and age columns, and parses overall survival months (`os_months`) and status (`os_status`).
- Appends driver mutations derived from MAF files.
- Log2-transforms expression data and aligns samples.

C. Riaz 2017 (`clean_riaz_2017`)

- Retains all samples (both pre and on-treatment) because `baseline_only=False` is set in the data cleaning script.
- Maps response outcomes to binary target `response_binary`.
- Resolves Riaz's mismatch by joining patient-level mutations with sample-level IDs in the clean mutations matrix.
- Transposes, log-transforms, and aligns all 107 samples.

D. TCGA-SKCM (`clean_tcga_skcm`)

- Merges sample and patient sheets and calls custom TCGA clean helpers (`clean_clinical_df`, `clean_rnaseq_df`).
- **Timeline Treatment Integration:** Checks for the existence of `data_timeline_treatment.txt` to parse and aggregate prior clinical treatments (e.g. pivoting key agents like Ipilimumab, Nivolumab, Pembrolizumab, Vemurafenib, Dabrafenib, Trametinib, and Interferon into binary flag columns).
- Filters out redundant and administrative clinical columns.

Cohort Attrition (Samples Lost at Each Step)

The following table summarizes the number of samples/patients retained and lost at each phase of the cleaning pipeline:

Cohort	Step	Starting N	Action / Filter	Lost	Retained N
Liu 2019	1	122	Raw patient-sample merge	-	122
	2	122	Response filter (keep CR/PR/PD; drop SD/MR/NaN)	18	104
	3	104	Expression matrix sample alignment	0	104
Hugo 2016	1	27	Raw patient-sample merge	-	27
	2	27	Response filter (keep CR/PR/PD; drop SD/MR/NaN)	0	27
	3	27	Expression matrix sample alignment	0	27
Riaz 2017	1	107	Raw patient-sample merge	-	107
	2	107	Response filter (keep CR/PR/PD; drop SD/MR/NaN)	43	64
	3	64	Expression matrix sample alignment	0	64
TCGA-SKCM	1	448	Raw patient-sample merge	-	448
	2	448	Cleaned intermediate clinical file size	0	448
	3	448	Patient deduplication in model analysis	6	442
	4	442	Survival validation cohort (drop invalid/missing OS)	15	427

3. Dataset Merging (`scripts/merge_datasets.py`)

After cleaning, the merge script combines all four cohorts into unified expression and clinical matrices. It produces **two** merged variants, each batch-corrected independently:

3a. Full Merge (`data/processed/merged/full/`)

- Concatenates **all** samples from TCGA-SKCM, Liu 2019, Hugo 2016, and Riaz 2017.
- Includes all treatment backgrounds (immunotherapy, chemotherapy, radiation, targeted therapy, and untreated TCGA patients).
- Suitable for pan-cohort analyses such as survival modelling and general biomarker discovery.

3b. Immunotherapy-Only Merge (`data/processed/merged/immunotherapy/`)

- Includes **all** samples from the three immunotherapy trial cohorts (Liu 2019, Hugo 2016, Riaz 2017), which exclusively enrolled patients on anti-PD-1 therapy.
- Additionally includes the subset of TCGA-SKCM patients flagged as having received immunotherapy (`TX_TYPE_IMMUNOTHERAPY == 1`).
- Suitable for immunotherapy-specific response prediction and immune biomarker analyses.

Shared Merging Workflow

Both variants follow the same processing steps:

1. **Gene Feature Alignment:** Intersects gene symbols across all four cleaned expression matrices, retaining only genes present in every cohort.
2. **Concatenation:** Row-wise concatenation of the selected expression and harmonised clinical dataframes.

- 3. **Zero-Variance Gene Removal:** Drops any genes with zero variance across the merged cohort to prevent division-by-zero errors during batch correction.
- 4. **pyCombat Batch Correction:** Applies parametric empirical Bayes batch-effect correction (ComBat) using the cohort label as the batch variable. The batch map is built dynamically from the cohorts present in each merge variant.
- 5. **Output:** Saves `expr_merged.csv` (batch-corrected expression) and `clin_merged.csv` (harmonised clinical metadata) to the respective output directory.

Harmonised Clinical Columns

Each sample in the merged clinical metadata includes the following standardised fields:

Column	Description
patient_id	Original patient identifier
cohort	Source dataset (TCGA, Liu_2019, Hugo_2016, Riaz_2017)
os_months	Overall survival in months
os_status	Overall survival status (1 = deceased, 0 = living)
response	Binary immunotherapy response (1 = CR/PR, 0 = PD; NaN for TCGA non-trial patients)
age	Age at diagnosis (where available)
sex	Sex (Male/Female/N/A)
specimen_type	Biopsy type (Primary/Metastatic/N/A)
immunotherapy	Whether the patient received immunotherapy (1 = yes, 0 = no)

4. Clinical & Genomic Characterisation Pipeline

Once the clean datasets are generated, the characterisation scripts analyze clinical and genomic variables across trials (Liu, Hugo, Riaz) and the TCGA reference cohort.

4.1. Clinical Characterisation

- `scripts/clinical_analysis/run_clinical_analysis.py` : Reads processed clinical data and generates stacked bar charts showing percentage response rates (CR/PR vs. PD) across studies, saved to `plots/clinical/response_distribution.png` and `plots/clinical/km_os_grid.png`.
- `scripts/exploratory_plots/run_response_distribution.py` : Generates cohort-level response breakdown bar charts.
- `scripts/exploratory_plots/run_waffle_chart.py` : Draws waffle charts representing absolute sample sizes and response status (1 block = 1 patient), saved to `plots/clinical/response_waffle_chart.png`.
- `scripts/exploratory_plots/run_response_km_curves.py` : Evaluates overall survival stratified by response (Responder vs. Non-Responder) in trials, generating Kaplan-Meier curves and Log-Rank tests saved to `plots/clinical/km_os_by_response.png`.
- `scripts/exploratory_plots/run_forest_plot.py` : Fits univariate logistic regression models for response across demographics and driver mutations. Generates a standardized forest plot (grey/red/blue color scheme) saved to `plots/clinical/forest_plot_odds_ratios.png`.
- `scripts/clinical_analysis/run_clinical_feature_selection.py` : Performs univariate and multivariate Cox proportional hazards modelling, Random Forest Gini feature selection, and Logistic Regression odds ratio evaluation across clinical phenotypes.

4.2. Genomic Characterisation

- `scripts/biomarkers/run_genomic_characterisation.py` : Evaluates baseline genomic properties of TCGA and trials:
 - Generates a comparison of driver mutations (BRAF , NRAS , NF1 , and Triple-WT) saved to `plots/genomic/mutation_frequencies.png`.
 - Plots pre-treatment TMB distributions (trial boxplots by response, TCGA log-normal histogram) saved to `plots/genomic/tmb_distribution.png`.
 - Plots a Spearman correlation matrix of somatic mutation and neoantigen loads in Liu 2019 saved to `plots/genomic/biomarker_correlation_heatmap.png`.
 - Generates Kaplan-Meier curves for TCGA overall survival by driver mutation subtype and TMB median-split saved to `plots/genomic/km_genomic_features.png`.
- `scripts/biomarkers/run_merged_comut_plot.py` : Aggregates clinical records and somatic mutations across the three trial studies to generate a pooled, sorted oncoplot ($N = 195$) showing driver/resistance gene states aligned with TMB, Response, Cohort source, and Sex. Saved to `plots/genomic/comut_landscape_merged.png`.
- `scripts/biomarkers/run_extended_biomarkers.py` : Evaluates advanced genomic biomarkers:
 - Correlates total predicted neoantigens with TMB in the pooled trials, generating a regression plot saved to `plots/biomarkers/extended_neoantigen_tmb.png`.
 - Correlates copy-number alterations (Aneuploidy Score in TCGA) and TMB against 5 continuous transcriptomic immune signatures in TCGA and pooled trials, saving the correlation matrix heatmap to `plots/biomarkers/extended_immune_correlations.png`.
 - Computes Kaplan-Meier survival curves in TCGA stratified by Aneuploidy Score, saved to `plots/biomarkers/extended_aneuploidy_survival.png`.
 - Trains cross-validated classifiers (Logistic Regression, Random Forest) on the pooled trial cohort ($N = 195$) to evaluate the predictive benefit of signatures, driver mutations, TMB, and pathway mutations.

5. Outputs Generated

The pipeline outputs processed data, figures, and reports to their respective directories:

Data Outputs (`data/processed/{study_name}/`)

- `expr_cleaned.csv` : Normalized and log2-transformed expression values (genes as columns, samples as rows).
- `clin_cleaned.csv` : Cleaned, standardized clinical metadata (patient demographic fields, survival timeline, response status, and driver mutation flags).

Merged Data Outputs (`data/processed/merged/`)

- `full/expr_merged.csv` : Batch-corrected expression matrix for the full merged cohort (all 4 datasets).
- `full/clin_merged.csv` : Harmonised clinical metadata for the full merged cohort.
- `full/merged_genomic.csv` : Unified genomic features (TMB, driver mutations, neoantigens) for the full merged cohort.
- `immunotherapy/expr_merged.csv` : Batch-corrected expression matrix for immunotherapy-treated patients only.
- `immunotherapy/clin_merged.csv` : Harmonised clinical metadata for immunotherapy-treated patients only.
- `immunotherapy/merged_genomic.csv` : Unified genomic features for immunotherapy-treated patients only.

Visualisation Outputs (`plots/`)

- **Clinical Characterisation**: Waffle charts, response rates, survival by response, and standardized univariate forest plots in `plots/clinical/`.
- **Genomic Characterisation**: Mutation landscapes, TMB distributions, correlation heatmaps, merged Co-Mutation oncoplots, and TCGA survival curves in `plots/genomic/` and `plots/biomarkers/`.
- **Model Performance**: ROC and PR curves in `plots/models/`.

Reporting & Inference Outputs (`reports/` and `predictor.py`)

- `predictor.py` : Standalone single-patient SVM response predictor CLI & Python API.
- `README.md` : Consolidated sitemap and 4-pillar documentation guide for the reports suite.
- `pillar-1-cohorts-and-preprocessing/` :
 - `cohort_characteristics_clinical.md` : Baseline report detailing clinical patient demographics, treatment histories, response distributions, survival curves, and forest plots.
 - `cohort_characteristics_genomic.md` : Baseline report detailing driver mutations, pathway mutations, TMB, neoantigens, immune signature correlations, Aneuploidy overall survival curves, and the merged CoMut oncoplot.
 - `batch_correction_report.md` : Evaluation report of technical batch effects (uncorrected PCA vs Z-score scaling vs ComBat) and cross-validation data leakage prevention.
- `pillar-2-clinical-subtyping/` :
 - `clinical_phenotyping_and_feature_selection.md` : Consolidated report detailing non-expression clinical feature importance (RF & Cox PH) and Ward's hierarchical subtype clustering.
- `pillar-3-transcriptomic-signatures/` :
 - `curated_signatures_report.md` : Consolidated report detailing gene signature implementation (IFN- γ , TIS, CYT, IMPRES, CD8, TCGA OS), signature scaling, biomarker orthogonality, 5-fold CV multimodal response predictor evaluation, and LOCO model validation.
- `pillar-4-out-of-cohort-benchmarks/` :
 - `model_evaluation_report.md` : Benchmark report detailing Leave-One-Cohort-Out (LOCO) cross-validation across all 5 classifiers (LR, RF, XGB, SVM, ElasticNet), ROC/PR curves, Youden thresholding, and downstream overall survival stratification.
 - `transcriptomic_feature_selection_results.md` : Benchmark report detailing response-based `SelectKBest` feature selection ($k = 20, 100, 200$) vs. signatures across LOCO cross-validation, TCGA Cox 20-gene survival signature derivation, and InterPro domain annotations.